

6 A linear transformation T of the x - y plane has an associated matrix $\mathbf{M}$, where $\mathbf{M} = \begin{pmatrix} \lambda & k \\ 1 & \lambda - k \end{pmatrix}$, and λ and k are real constants.

(a) You are given that $\det \mathbf{M} > 0$ for all values of λ .

(i) Find the range of possible values of k . [3]

(ii) What is the significance of the condition $\det \mathbf{M} > 0$ for the transformation T ? [1]

For the remainder of this question, take $k = -2$.

(b) Determine whether there are any lines through the origin that are invariant lines for the transformation T . [4]

(c) The transformation T is applied to a triangle with area 3 units². The area of the resulting image triangle is 15 units².
Find the possible values of λ . [3]